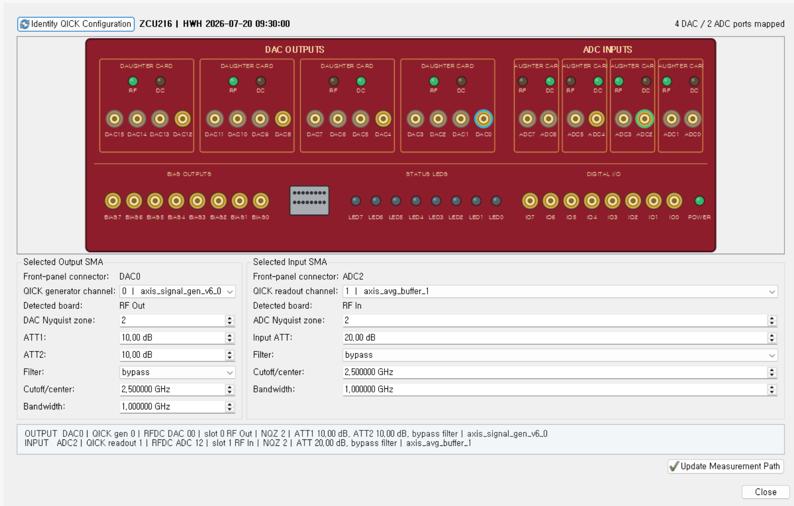
PulseGenerator GUI

User Guide for QICK-Based Pulse and RF Measurements

AWG tuning, RF output/readout, FIR-DDR capture, stability diagrams, S-parameters, calibration, and QCoDeS storage

Audience: experimentalists familiar with RF measurements who are new to QICK

Version 1.0 | 20 July 2026



1. What This Application Controls

Control travels left to right; captured FIR I/Q and metadata return to the host.

QICK in one paragraph

QICK combines a software-defined experiment description with FPGA data paths. The tProcessor dispatches timestamped commands. Generator IP drives RFDC DAC channels, readout IP processes RFDC ADC channels, and custom FIR/DDR logic stores long traces. The GUI builds this configuration and keeps channel routing, RF-board settings, pulse timing, and database metadata together.

- The GUI does not stream every waveform point from the PC in real time.
- Pulse commands and sweep parameters are compiled before execution.
- FIR I/Q data returns after FPGA processing and DDR capture.

Major hardware terms

- **RFDC**: integrated DAC and ADC tiles on the ZCU216.
- **tProcessor**: deterministic timing engine that issues generator and trigger commands.
- **Fabric clock**: PL clock used by AXIS generators and custom IP.
- **HWH**: hardware metadata describing instantiated IP and signal connectivity.
- **FIR-DDR**: filtered, decimated capture path that writes I/Q samples into PL DDR.

Firmware assumption
This guide targets the qstl_awg_tuning_fir image and its custom AWG tuning and FIR-DDR path. A different bitstream can expose a different channel map or feature set.

2. Installation and First Connection

Recommended host environment

Use the repository requirements in a dedicated Conda environment. The GUI needs PyQt5, pyqtgraph, NumPy, QCoDeS, Pyro, and the matching local QICK package.

Typical launch

```
conda activate qick_test
python DCWaveformGeneratorGUI/DCWaveform_Generator.py
```

QICK server

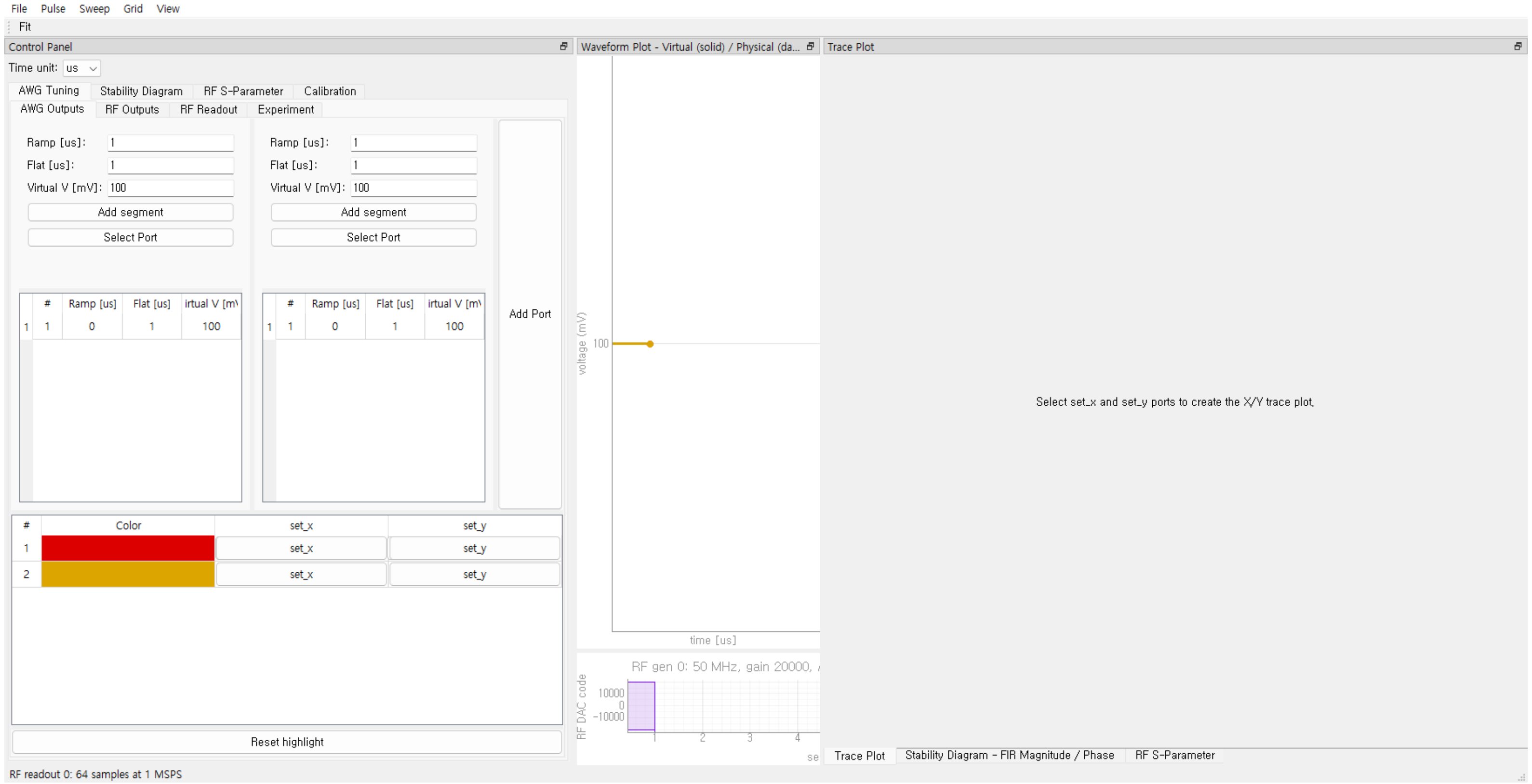
Start the Pyro nameserver and QICK server on the ZCU216, then enter the host, nameserver port, and proxy name in AWG Tuning > Experiment.

Before energizing a device

1. **Load firmware** - Confirm the intended BIT/HWH pair is active.
2. **Identify configuration** - Open the Front Panel and read live HWH routing.
3. **Verify daughter cards** - Confirm RF_Out, DC_Out, RF_In, or DC_In for each connector.
4. **Set RF protection** - Start with high attenuation and a conservative DAC gain.
5. **Dry run** - Use Show Program before the first hardware execution.

Clock defaults
The GUI defaults the AWG fabric clock and tProcessor time base to 300 MHz. Keep these values cons

3. Main Window and Settings Files



Dock layout

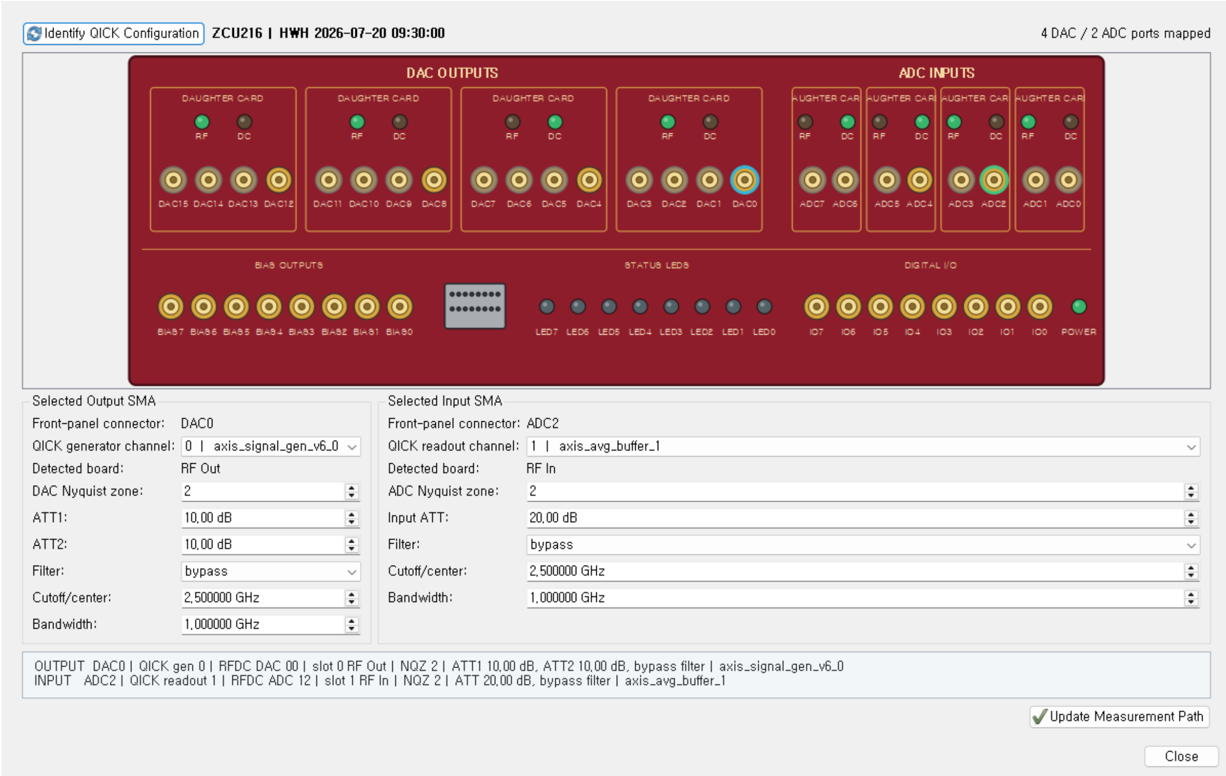
- The left Control Panel contains AWG Tuning, Stability Diagram, RF S-Parameter, and Calibration tabs.
- Waveform, trace, stability, and S-parameter plots are movable dock windows.
- Drag dock dividers to resize; long control pages use vertical scrolling.

JSON project files

- File > Save stores waveforms, sweeps, cross-capacitance, RF settings, databases, and display preferences.
- Older files receive default values for newly added fields, including Nyquist zones and Stability DB path.
- Saving again upgrades the schema to the current version.

4. HWH-Backed Front Panel

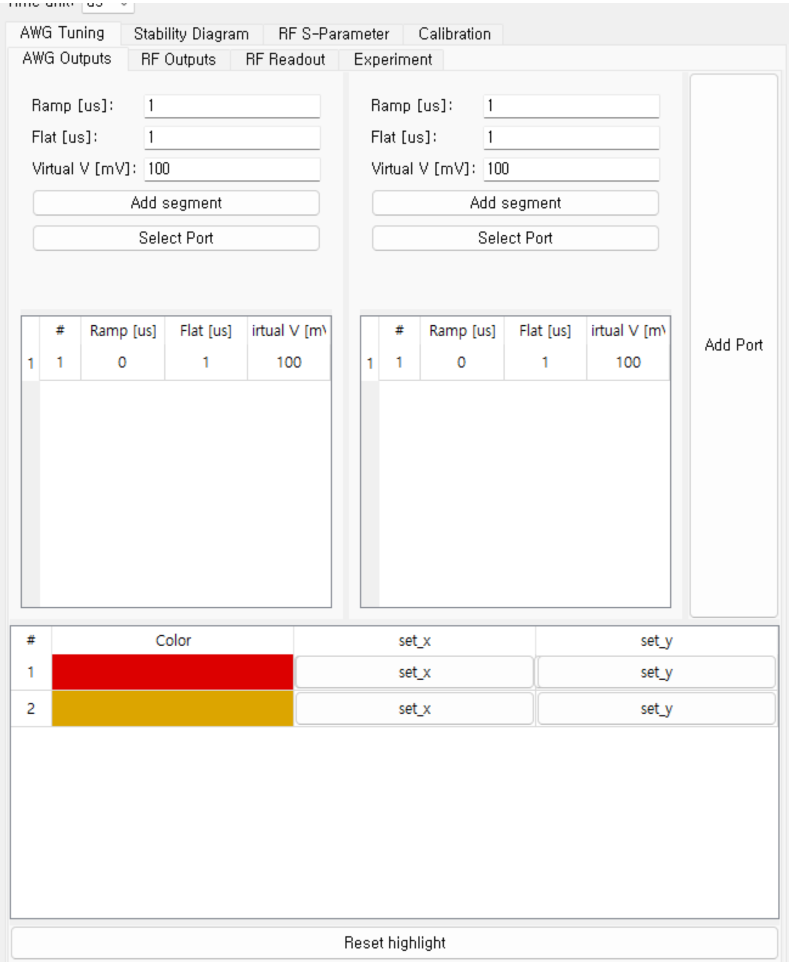
Use the graphical panel as the authoritative RF path editor.



- Click **Identify QICK Configuration** to read HWH routing and detected daughter-card types.
- Select a DAC or ADC SMA directly on the panel. The mapped QICK generator/readout channel appears below.
- For RF boards, configure attenuators and programmable filters. DC cards hide controls that do not exist.
- Set DAC and ADC Nyquist zones independently to 1 or 2.
- The summary line lists physical connector, RFDC tile/block, board slot, NQZ, attenuation, filter, and IP path.
- Click **Update Measurement Path** to propagate the committed path to AWG Tuning, Stability, S-Parameter, and Calibration.

Nyquist zones
Changing NQZ changes the RFDC reconstruction or sampling-zone configuration, not the requested logical sweep frequency. Confirm the analog band and anti-alias filtering before using zone 2.

5. AWG Waveform Authoring



Building a sequence

- Each AWG output is a virtual electrode waveform composed of SET and RAMP segments.
- Add a segment with ramp duration, flat duration, and target voltage.
- Right-click a segment to edit, remove, or assign a voltage sweep.
- The waveform plot shows virtual traces as solid lines and physical outputs as dashed lines.

Cross-capacitance

The cross-capacitance matrix maps virtual electrodes to physical DAC voltages. Review both views before execution, especially when multiple virtual gates move together.

Grid and snapping

Grid settings control visible time/voltage spacing and optional point snapping. Time units can be ns, us, or ms without changing stored physical time.

6. Sweeps and Bias-T Compensation

Hardware voltage sweeps

- A sweep is assigned to a specific output and SET segment.
- Specify start, stop, and point count in physical voltage units.
- Adjacent ramps are recalculated automatically when the swept SET level changes.
- Multiple sweeps form a Cartesian product; repetitions are nested inside each sweep point.

tProcessor implementation

Sweep values are advanced using tProcessor registers and loops. This avoids emitting a separate host-generated program for every voltage point and preserves deterministic timing.

DC compensation

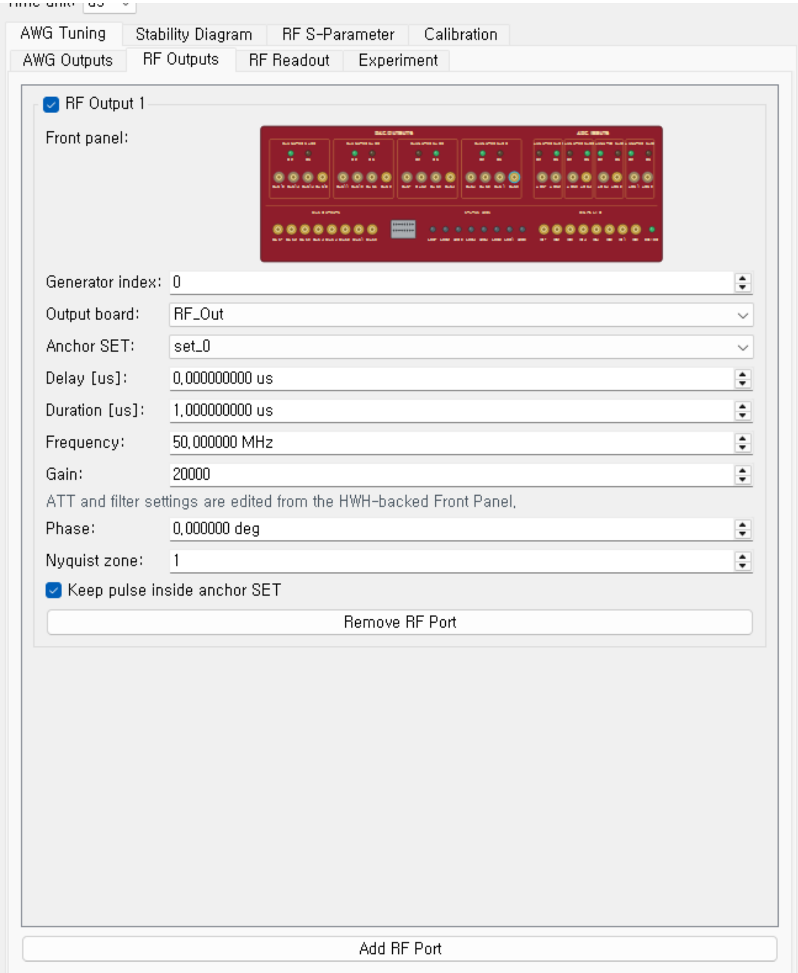
A compensation voltage of opposite area is appended after the main waveform. In fixed-voltage mode, duration is calculated. In fixed-duration mode, voltage is calculated.

Filter compensation

For flat segments, a user-supplied time constant tau generates a target/tau slew-rate correction. The sequence compiler calculates correction commands before the shot; it does not allocate a waveform-sized tProcessor memory table.

Safety
Compensation can increase peak voltage or sequence duration. Verify physical traces after cross-capa

7. RF Output Channels



RF pulse placement

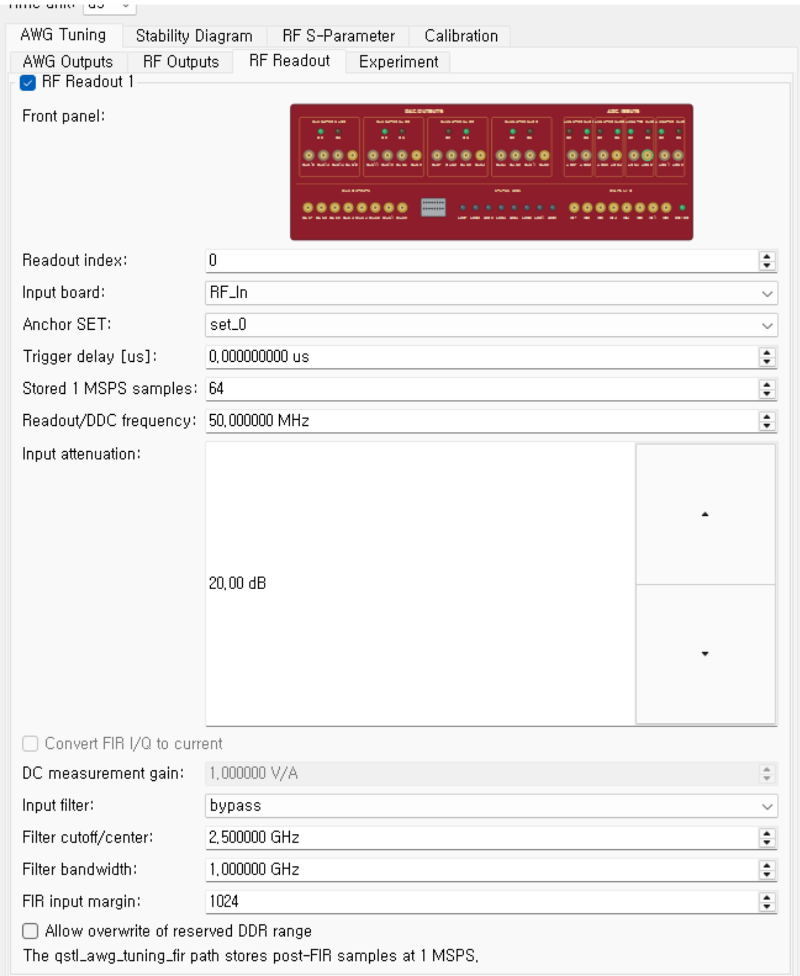
- Enable an RF output and select generator index, anchor SET, delay, duration, frequency, gain, phase, and NQZ.
- Keep pulse inside anchor SET rejects schedules that extend outside the selected flat segment.
- The generator runs in periodic mode and receives a timed zero/stop command, allowing durations beyond one pulse-register length field.

Centralized board controls

ATT1, ATT2, board type, and output filter are intentionally edited from the HWH-backed Front Panel. This prevents conflicting duplicate values in the RF Output and S-Parameter tabs.

Gain limit
The software clamps or rejects settings above the configured maximum QICK gain. Start measuremen

8. RF Readout and FIR-DDR Capture



Readout setup

- Select the readout index, board type, anchor SET, trigger delay, stored sample count, and DDC frequency.
- RF_In uses input attenuation and an RF filter. DC_In uses programmable input gain and can enable DC current conversion.
- The front-panel dialog also controls the ADC Nyquist zone.

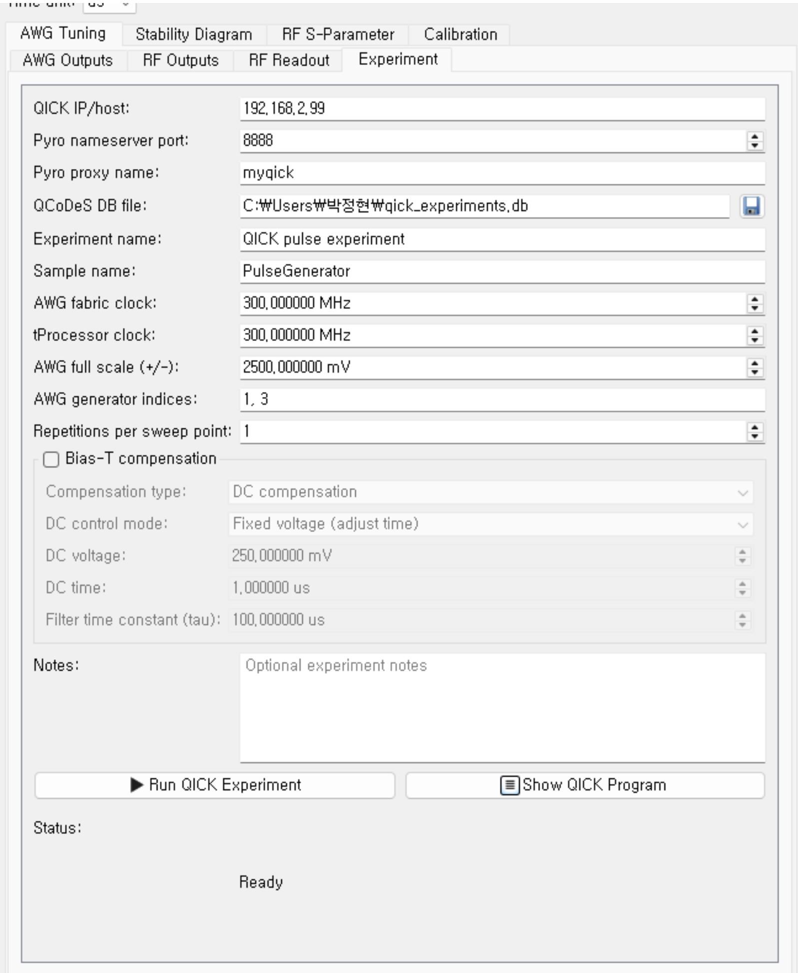
What is stored

The qstl_awg_tuning_fir design continuously filters the ADC stream, aligns decimation phase to the trigger, and writes post-FIR I/Q into PL DDR. The nominal stored rate is 1 MSPS for the deployed 300-to-1 decimation path.

Margins and overwrite

FIR input margin covers filter warm-up. DDR overwrite must be enabled only when the reserved capture range is known to be safe.

9. Running an AWG Tuning Experiment



Connection

- Enter QICK IP/host, Pyro nameserver port, and proxy name.
- Set AWG fabric clock, tProcessor clock, full-scale voltage, and AWG generator indices.

Execution

1. Show Program - Inspect generated QICK assembly and timing before hardware execution.

2. Choose database - Select a local QCoDeS DB when practical.

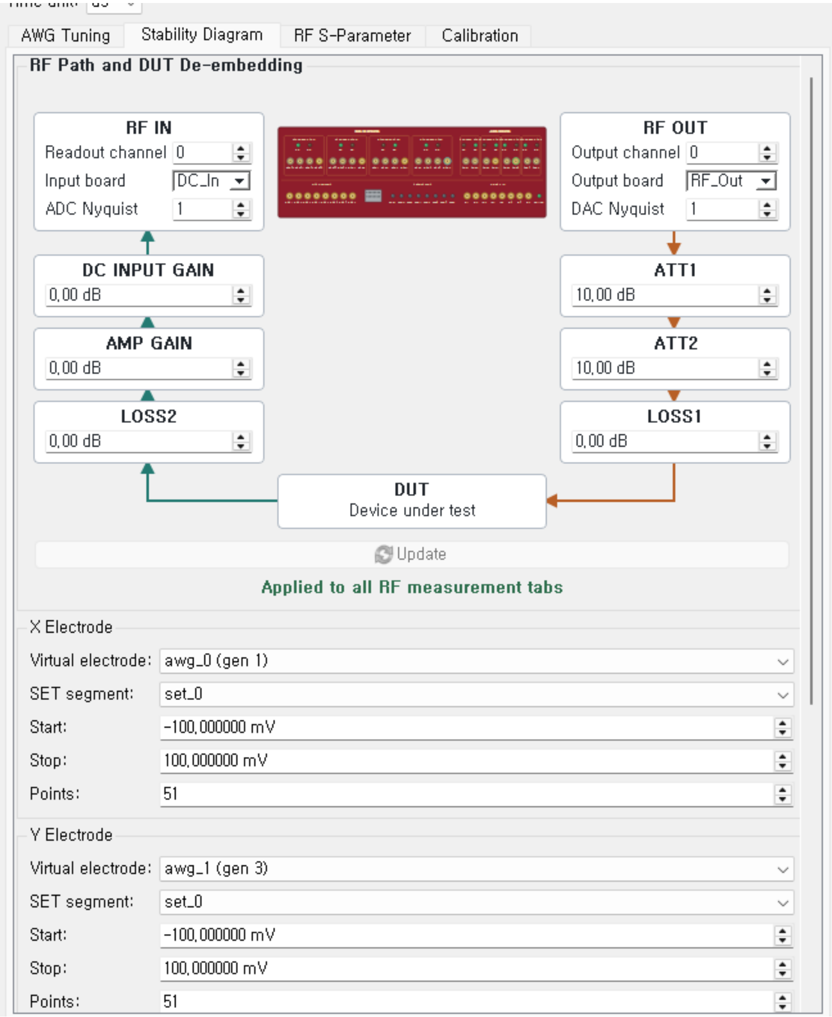
3. Run QICK Experiment - The progress indicator follows sweep and repetition completion.

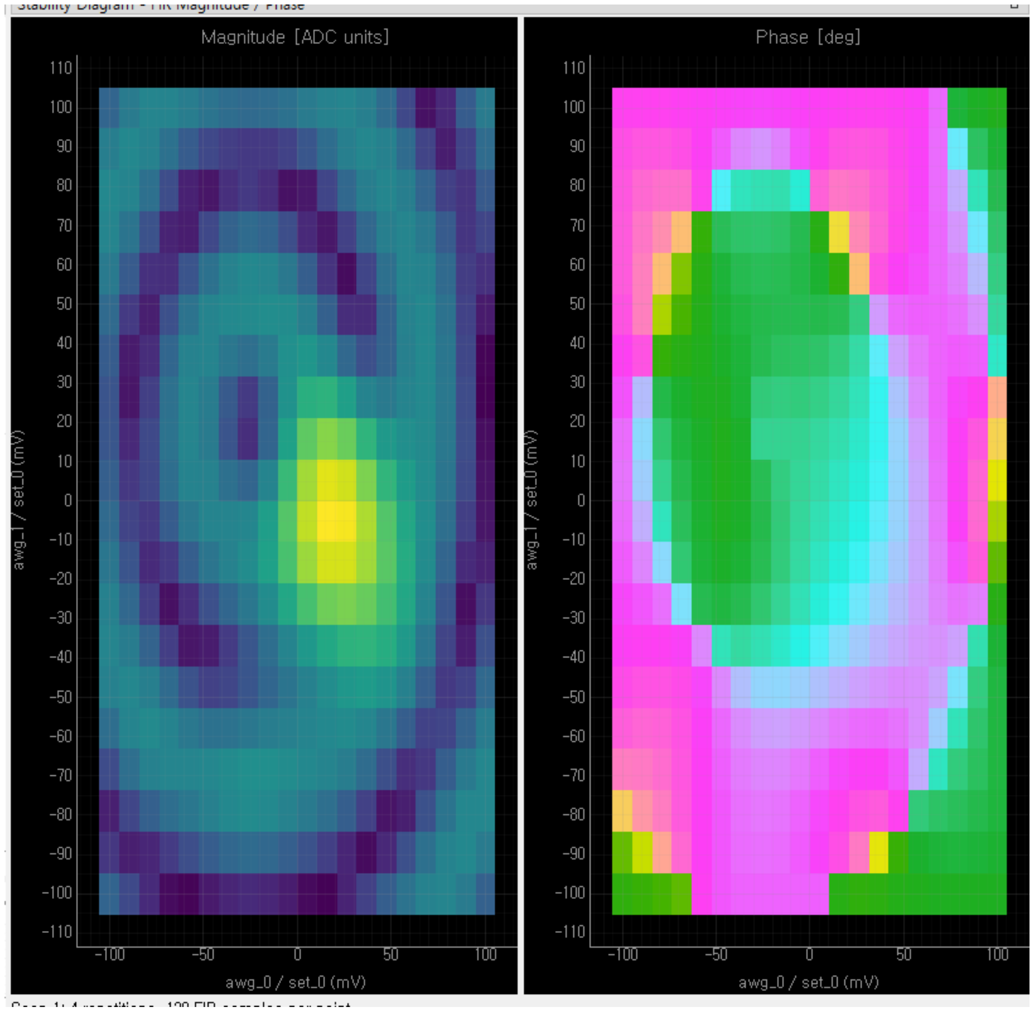
4. Inspect results - I and Q are stored as complete trace arrays per shot; magnitude and phase can be derived later.

Database strategy

For large runs, write to a local SSD, checkpoint SQLite WAL at completion, and then copy the DB to synchronized storage. Avoid one SQL row per individual sample.

10. Stability Diagram

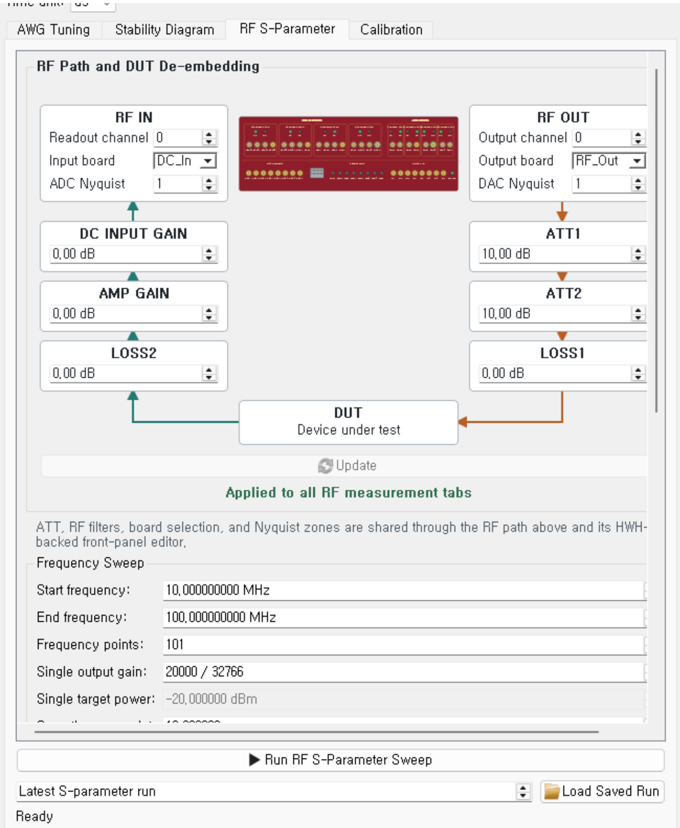


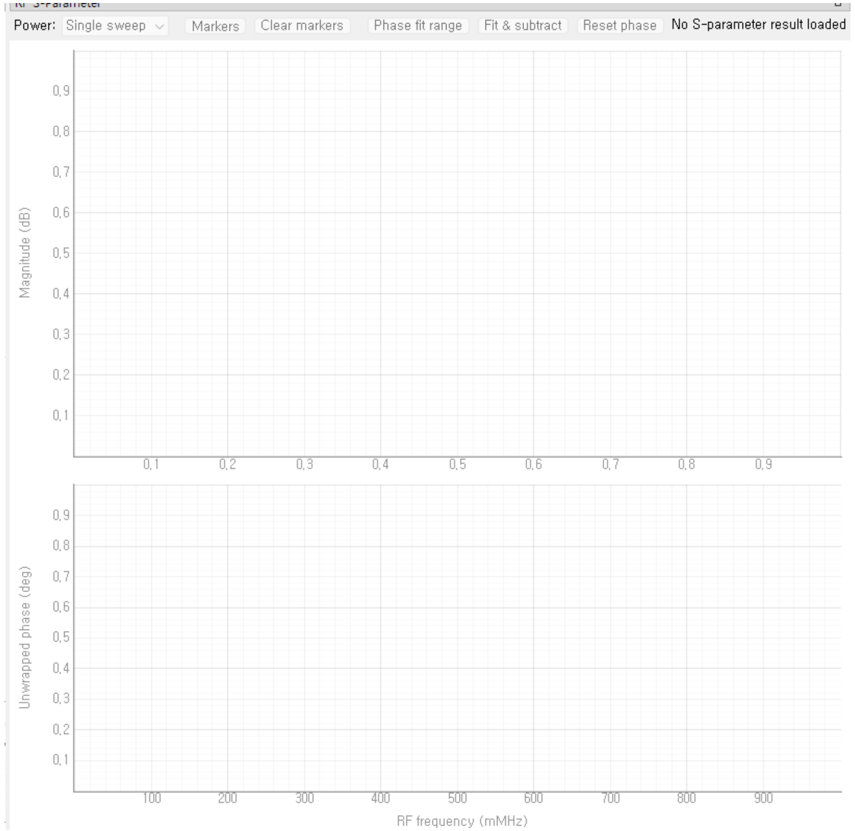


- The shared RF path at the top selects output/readout channels, boards, NQZ, attenuation, and de-embedding terms.
- X and Y electrodes each select an AWG output, SET segment, voltage range, and point count.

- Start repeats complete scans continuously without database writes. Stop exits after the active scan.
- Single Shot & Save writes one complete diagram to the DB selected in this tab.
- Magnitude and phase are calculated from coherent mean I/Q at every Cartesian point.

11. RF S-Parameter Sweep





- Frequency sweep points execute in hardware. Optional power points are an outer software sweep.
- For each point, FIR-DDR I/Q is read and coherently averaged to one complex response.
- Raw magnitude uses 20*log10(hypot(I,Q)); phase uses atan2(Q,I) and can be unwrapped.

- Markers show frequency and trace value on hover.
- Fit & subtract removes a selected linear phase trend.
- For power sweeps, the Power selector displays one trace or the requested power slice.
- The S-parameter DB is independent of the AWG Experiment DB.

12. RF Path Correction and Calibration

Path terms

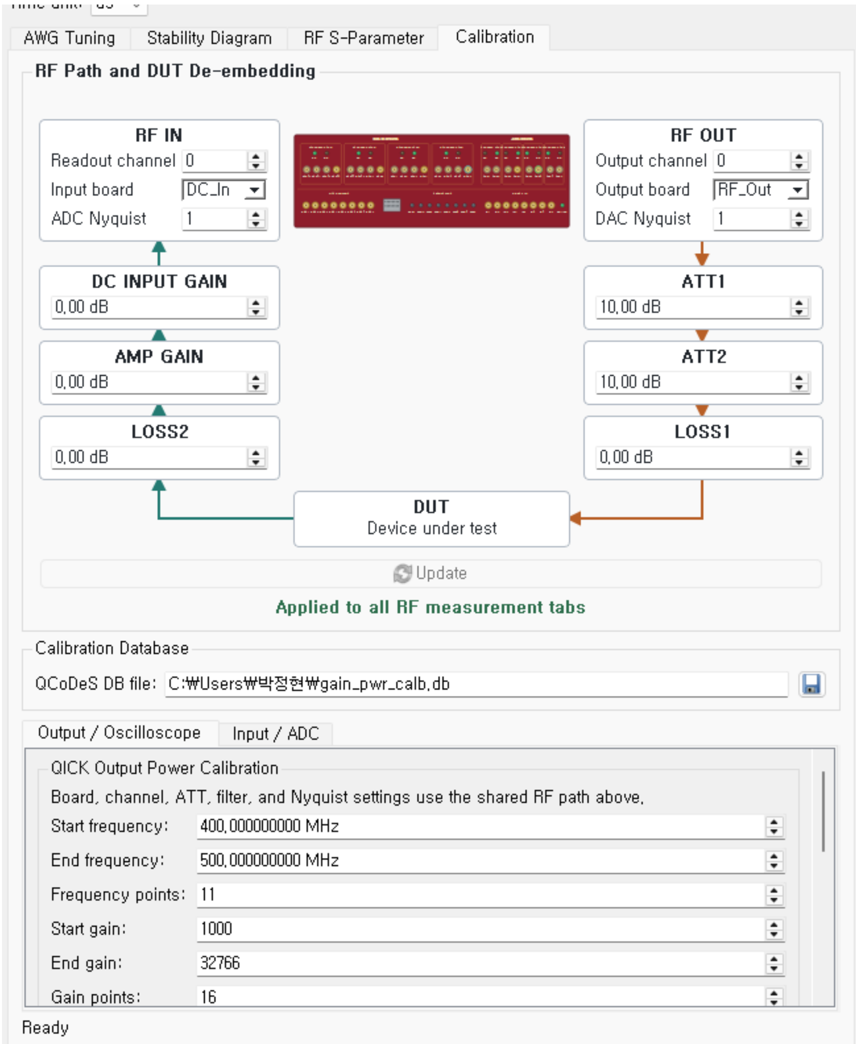
- P_DUT,in = P_RF,OUT - LOSS1
- P_DUT,out = P_RF,IN + LOSS2 - AMP_GAIN
- S21 = P_DUT,out - P_DUT,in

Board-aware correction

RF_Out contributes ATT1 and ATT2; RF_In contributes input attenuation. DC cards omit attenuators that are not physically present. Calibration lookup requires matching board type, overlapping frequency range, and compatible input attenuation.

Frequency response compensation

A calibrated output-response curve supplies one gain code per frequency. For more than 4096 frequency points, the table is compressed to DMEM capacity and nearest calibrated entries are reused.



13. Calibration Workflows

Output / oscilloscope calibration

1. Select path - Choose the RF output SMA and board in the shared path.

2. Define axes - Set frequency and DAC gain ranges.

3. Connect scope - Enter VISA resource, input channel, FFT function, span, averaging, and settle time.

4. Run - Measured power is stored with ATT, filter, board, frequency, gain, and instrument metadata.

Input / ADC calibration

1. Select path - Choose calibrated output and ADC input using the same front-panel model.

2. Acquire - Sweep frequency and gain while FIR-DDR returns I/Q.

3. Fit - Map calibrated input power to measured ADC magnitude.

4. Inspect - Plot input power versus ADC magnitude with linear or logarithmic axes.

Attenuation matching
When applying an input calibration, the GUI warns if the measurement attenuation differs. Select a manual Run ID or the latest calibration with the same input attenuation.

Physical S21
When compatible output and input calibrations are available, the plotted response uses calibrated DUT input and output powers. Without calibration, the trace is a relative ADC response, not an absolute VNA-grade S21.

14. QCoDeS Data Layout

Per-shot coordinates

- Sweep coordinates identify output and SET segment, not generic sweep0/sweep1 labels.
- repetition_index identifies the repeated shot inside one Cartesian sweep point.
- I trace and Q trace are stored as full-length one-dimensional arrays.

AWG vertices

- Each AWG channel stores virtual and physical vertex traces separately.
- Vertex time is reconstructed from paired time and voltage arrays; it is not exposed as a sweep selector.

Metadata

- BIT/HWH identity, clocks, channel map, board types, NQZ, attenuation, filters, and RF settings.
- Pulse sequence, cross-capacitance matrix, compensation mode, sweep definition, and repetitions.
- FIR sample rate, sample count, scan timing, calibration Run IDs, and de-embedding terms.

Plottr-inspectr

Select I trace or Q trace as the dependent array and sample index as its array axis. Use sweep coordinates and repetition_index to choose or compare shots.

Database size
Array-per-trace storage avoids millions of scalar SQL rows. Total size still scales with Cartesian points x repetitions x samples x two channels, so estimate storage before a large run.

15. Troubleshooting and Operating Checklist

Common problems

- **Channel appears wrong**: re-run Identify QICK Configuration and confirm HWH matches BIT.
- **First trace is noisy**: verify arm-before-trigger order, readout warm-up, and trigger offset. Do not assume DDR is the only source.
- **Timing scales differ**: verify the manual tProcessor and fabric clock values against deployed firmware.
- **Attenuator does not respond**: confirm daughter-card seating and detected board type before changing software.
- **No Numpy._core through Pyro**: return plain lists/scalars across RPC or align NumPy serialization versions.

Pre-run checklist

1. **Firmware** - Correct BIT and HWH loaded.
2. **Routing** - Physical SMA, logical channel, board type, and NQZ confirmed.
3. **Protection** - ATT/filter/gain and voltage full-scale reviewed.
4. **Timing** - Pulse, readout, FIR margin, and compensation windows checked.
5. **Storage** - DB path, free space, sweep size, and repetition count checked.
6. **Dry run** - Generated program inspected before energizing the DUT.

Support information
When reporting a problem, include the saved JSON, software commit, BIT/HWH pair, HWH channel summary, QCoDeS Run ID, complete traceback, and the exact settings that produced the issue.